LEARNER GUIDE

# Angles in Polygons

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps.
*Connection before curriculum. Always.*

## What this lesson is about

The angles inside a polygon are not random — they are decided by the number of sides. In this lesson you discover why: any polygon splits into triangles, and each triangle holds 180°. You will also meet a surprise that never changes — walk once around any polygon and you turn through exactly one full turn, 360°. The question you are exploring is: what decides the angles inside a polygon, and what stays the same however many sides it has?

## What you are learning to notice

- That a polygon splits into triangles — always two fewer than its number of sides — so its interior angles add to $(n - 2) \times 180°$.
- How to find each interior angle of a regular polygon by sharing that total between its corners.
- That the exterior angles of any polygon always add to 360°, however many sides it has.
- That the interior and exterior angle at a corner add to 180°, so knowing one gives you the other.

## Moving through it, one step at a time

Use the Polygon Explorer: choose the number of sides, draw the triangulation lines from one corner, count the triangles, and fill in the table — the pattern appears before you write the formula. Take the Exterior-Angle Walk to watch the turns add up to a full circle. Then investigate which regular polygons tessellate, and reason out why. There is no rush: try a prediction before you check, and use any support that helps.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the angle sum of a triangle, angles on a straight line and around a point, and naming polygons), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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